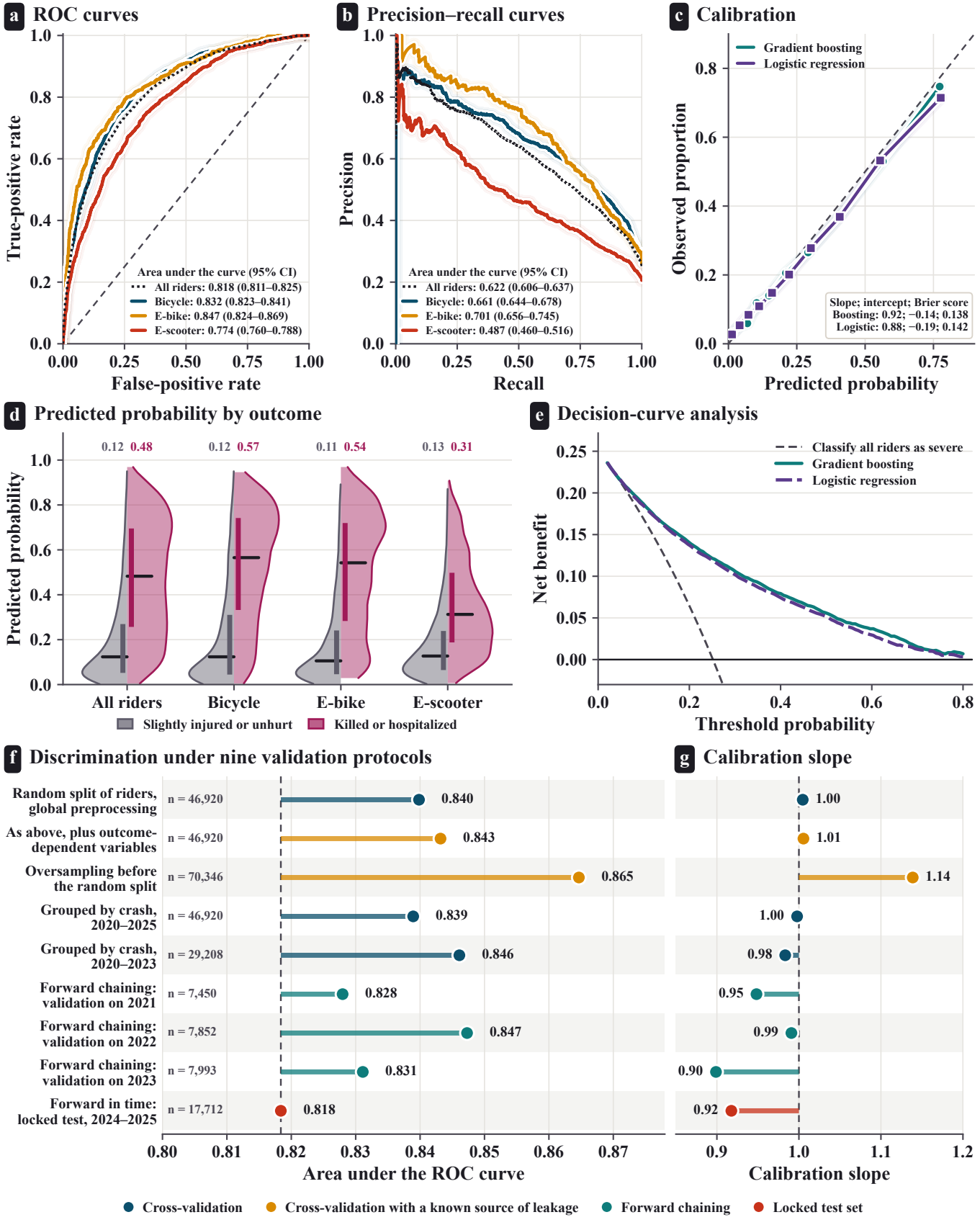

a ROC curves
True-positive rate
False-positive rate
Area under the curve (95% CI)
All riders: 0.818 (0.811–0.825)
Bicycle: 0.832 (0.823–0.841)
E-bike: 0.847 (0.824–0.869)
E-scooter: 0.774 (0.760–0.788)
b Precision–recall curves
Precision
Recall
Area under the curve (95% CI)
All riders: 0.622 (0.606–0.637)
Bicycle: 0.661 (0.644–0.678)
E-bike: 0.701 (0.656–0.745)
E-scooter: 0.487 (0.460–0.516)
c Calibration
Gradient boosting
Logistic regression
Observed proportion
Predicted probability
Slope; intercept; Brier score
Boosting: 0.92; −0.14; 0.138
Logistic: 0.88; −0.19; 0.142
d Predicted probability by outcome
Predicted probability
0.12 0.48
0.12 0.57
0.11 0.54
0.13 0.31
All riders
Bicycle
E-bike
E-scooter
Slightly injured or unhurt
Killed or hospitalized
e Decision-curve analysis
Classify all riders as severe
Gradient boosting
Logistic regression
Net benefit
Threshold probability
f Discrimination under nine validation protocols
Random split of riders, global preprocessing
n = 46,920
0.840
As above, plus outcome-dependent variables
n = 46,920
0.843
Oversampling before the random split
n = 70,346
0.865
Grouped by crash, 2020–2025
n = 46,920
0.839
Grouped by crash, 2020–2023
n = 29,208
0.846
Forward chaining: validation on 2021
n = 7,450
0.828
Forward chaining: validation on 2022
n = 7,852
0.847
Forward chaining: validation on 2023
n = 7,993
0.831
Forward in time: locked test, 2024–2025
n = 17,712
0.818
Area under the ROC curve
g Calibration slope
1.00
1.01
1.14
1.00
0.98
0.95
0.99
0.90
0.92
Calibration slope
Cross-validation
Cross-validation with a known source of leakage
Forward chaining
Locked test set